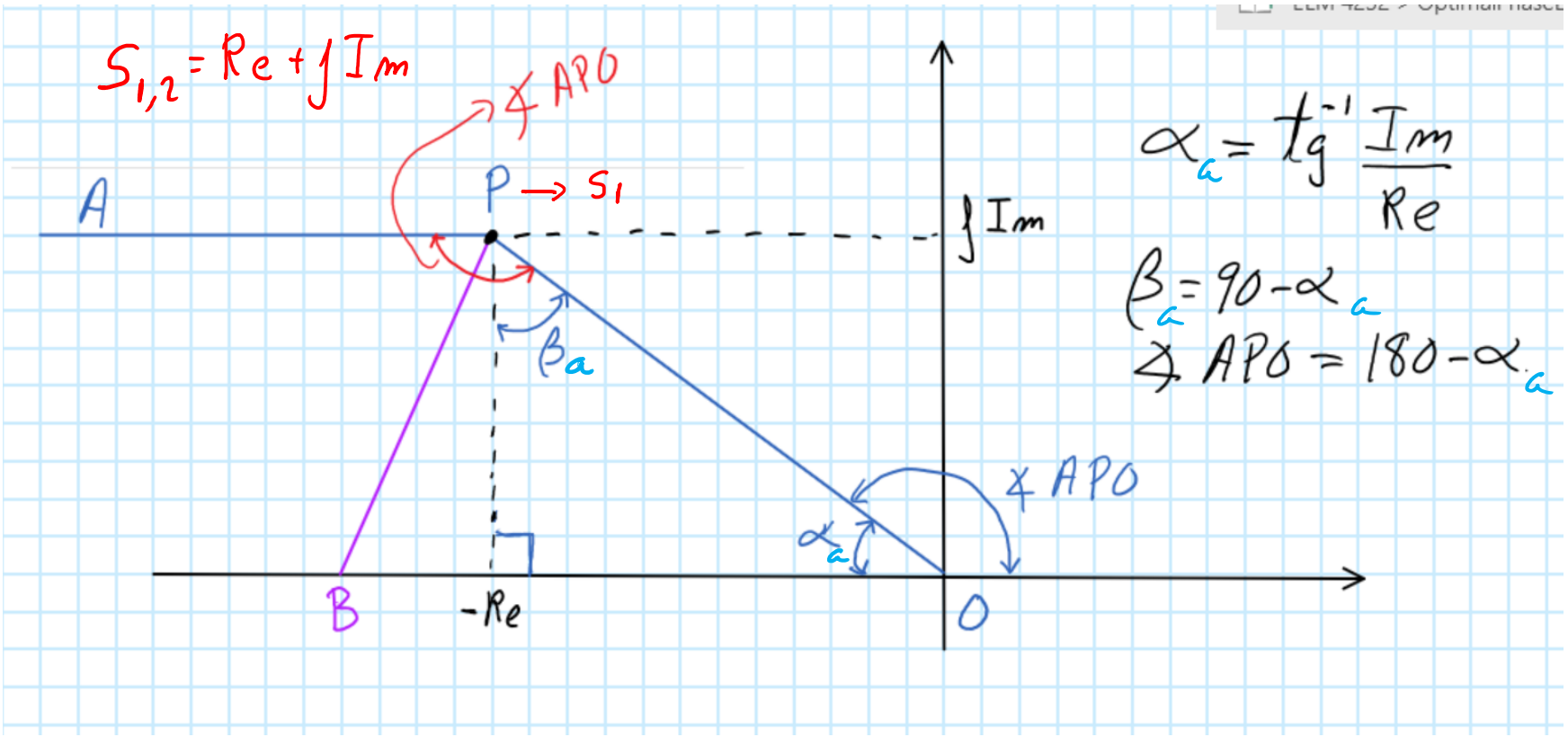
Optimal Phase Lead Controller

Geometric Approach

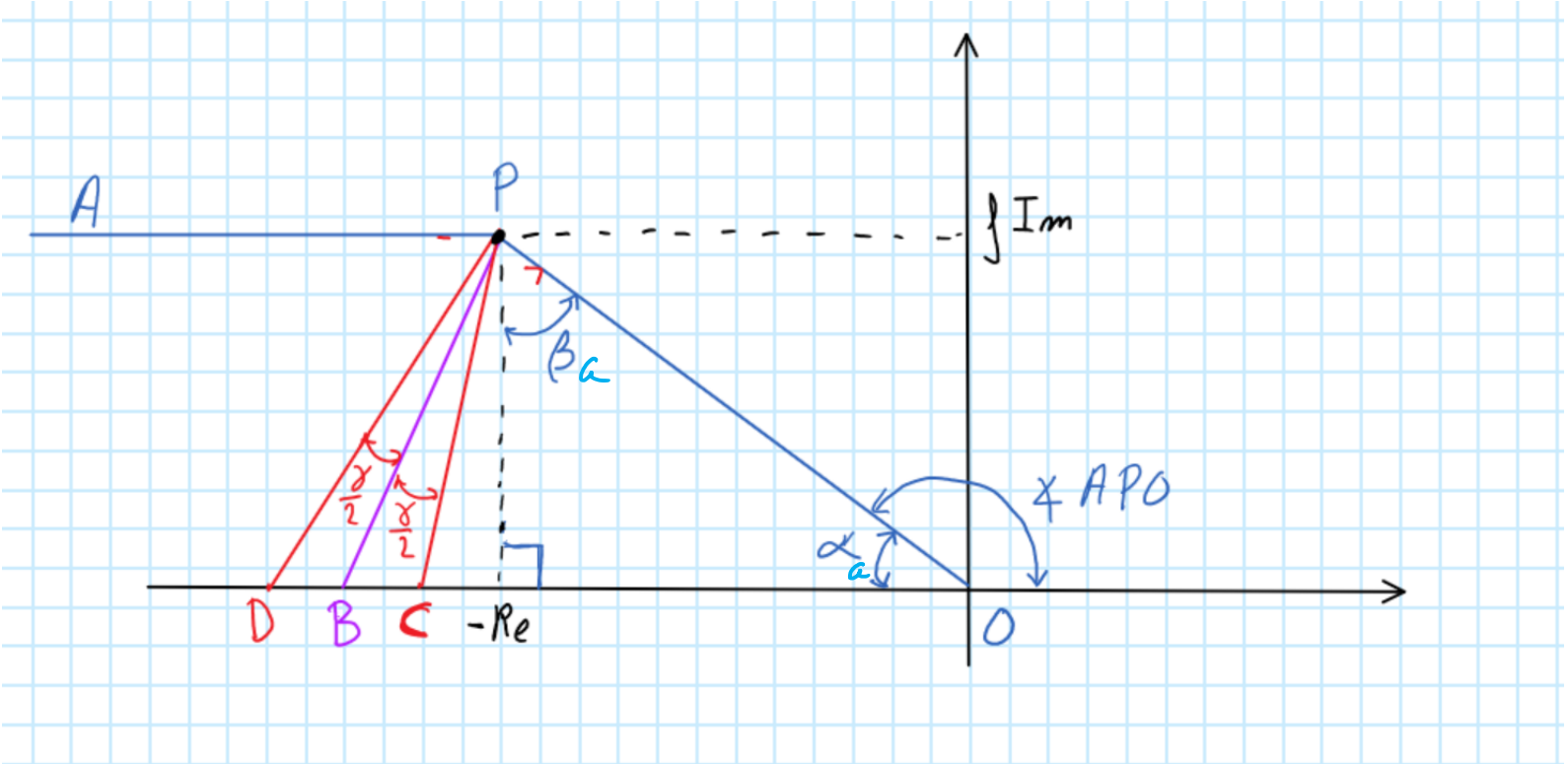
Step 1

1. Draw the lines AP and PO (you know the point P, desired pole).
2. Draw the line PB, such that PB bisects $\measuredangle APO \Rightarrow \measuredangle APB = \measuredangle BPO = \measuredangle APO/2$



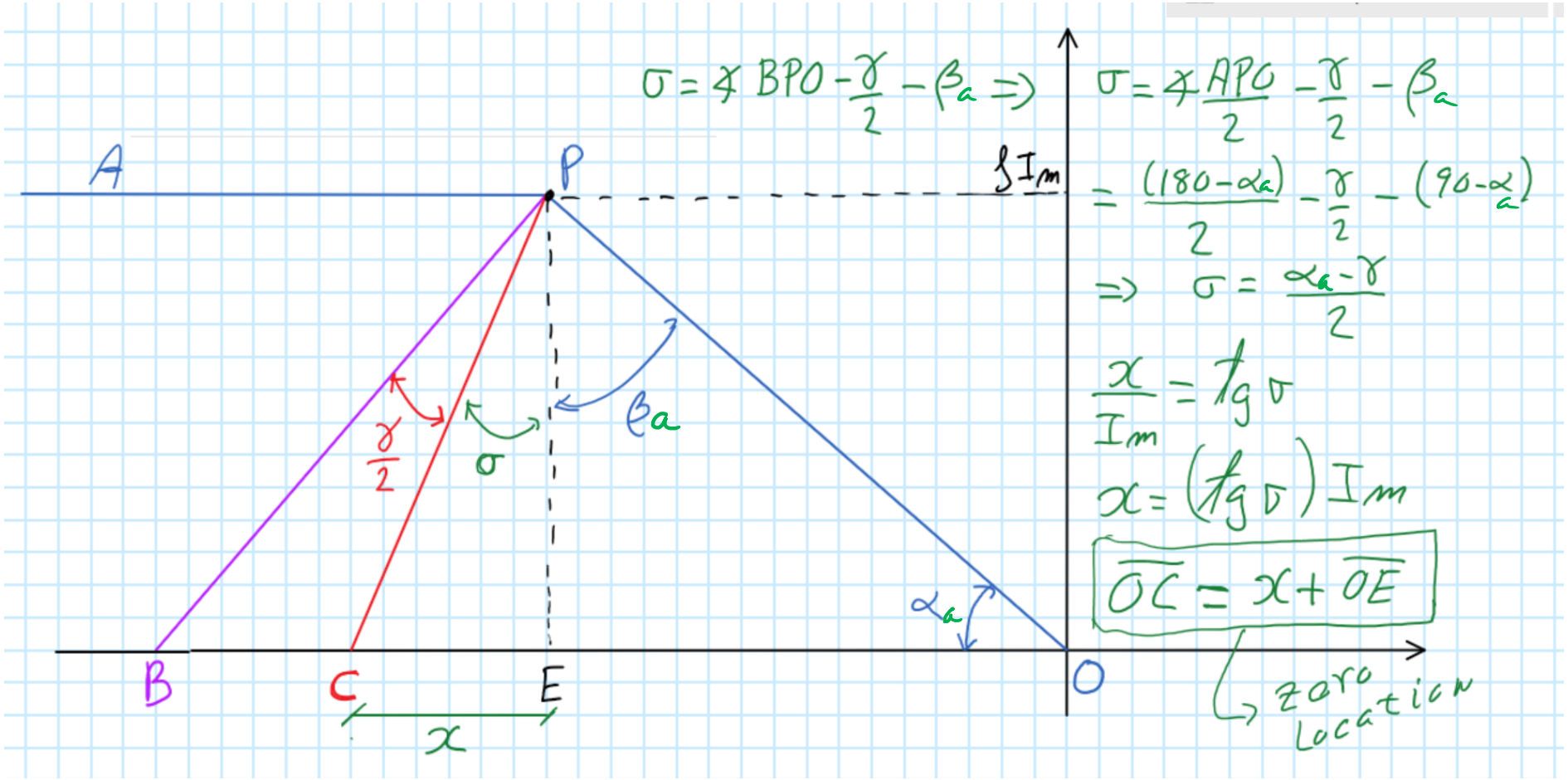
Step 2

3. Draw the lines PC and PD such that $\measuredangle DPB = \measuredangle BPC = \frac{\gamma}{2}$, where γ is the angle deficiency that is known. C and D are the locations of the lead compensator zero and pole, respectively.



Step 3

4. Work to find the zero first. Notice that – Re has been replaced by E.

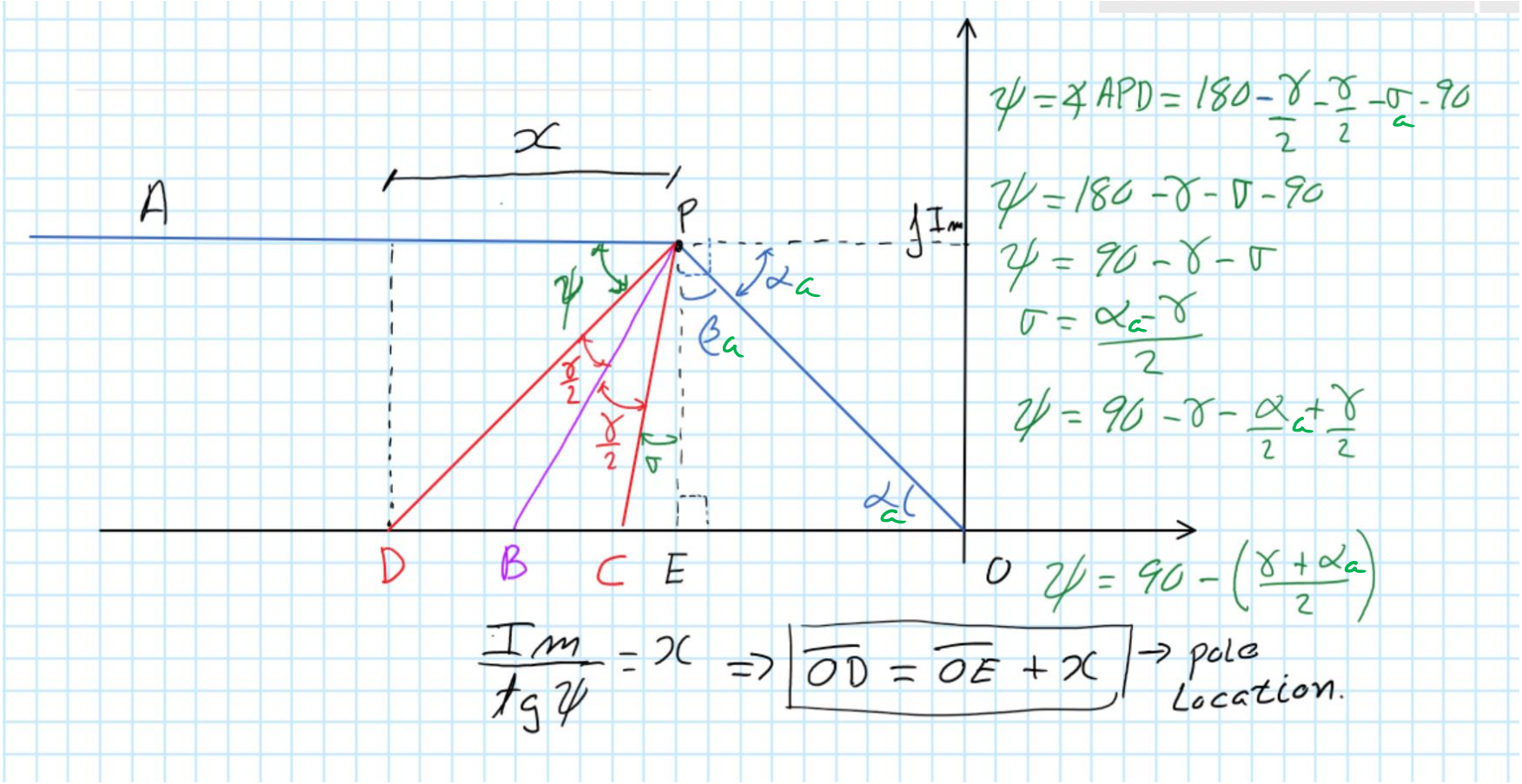


Step 3 … Conclusion for finding the zero (Follow these steps to find the zero)

- To find the zero location, you need to find the angle σ indicated in the figure of Slide 4, which is a simple expression that depends on α_a and γ (angle deficiency). If σ is positive, the point C is on the left of E; if σ is negative, the point C is on the right of E.
- Next, find x. x is negative if σ is negative.
- Finally, you find the zero location by finding $\overline{OC}$.

Step 4

- Find the pole location. Find the angle ψ, given the formula below



Conclusion for finding the pole (Follow these steps to find the pole)

- Find the angle ψ (Slide 6).
- Find x (Slide 6)
- Finally, you find the pole by finding $\overline{OD}$ (Slide 6)